# Mitoprep Answer Key 3

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Q1) **Explanation:**

**Hint:** Area $= \int F dt$

**Step:** Find the change in momentum of the particle.
The change in momentum of the particle can be given by the area under the curve $F$(force) and $T$(time).
The change in linear momentum in the given interval of time = Area below the $F-T$ graph in that interval
The change in momentum $= 2 \times \frac{6}{2} - 2 \times 3 + 4 \times 3 = 12$ Ns
Hence, option **(3)** is the correct answer.

Q2) **Explanation:**

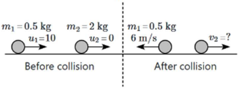
**Hint:** Energy is given by $E = P \times t$

**Step:** Find the energy radiated by the body.
$E = P \times t = 100 \times 1000 \times 60 \times 60 = 36 \times 10^7$ J
Hence, option (2) is the correct answer.

Q3) **Explanation:**

**Hint:** Apply the conservation of momentum.

**Step:** Find the speed of the $2.0$ kg ball after collision.
The initial and the final velocities of the ball before and after the collision are shown in the figure below.




Let the velocity of the $2.0$ kg ball after collision is $v_2$ m/s.
Given:
$m_1 = 0.5$ kg, $m_2 = 2.0$ kg, $u_1 = 10$ m/s, $u_2 = 0$ m/s, $v_1 = -6$ m/s
By applying the conservation of linear momentum, we get;
$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$
$\Rightarrow 0.5 \times 10 + 2 \times 0 = 0.5 \times (-6) + 2 \times v_2$
$v_2 = \frac{8}{2} = 4$ m/s
Therefore, the speed of the $2.0$ kg ball after collision is $4$ m/s.
Hence, option (3) is the correct answer.

Q4) **Explanation:**

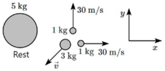
**Hint:** When the block is lifted off, $N = 0$

**Step:** Find the minimum value of $M$.
The tension in the string for the minimum mass condition is given by;
$T = mg$
For mass $M$ to just begin lifting off the table, the tension $T$ $(T = mg)$ must overcome the force due to the weight of mass $M$ i.e.,
$mg = Mg \sin 60^\circ$
$\Rightarrow M = \frac{m}{\sin 60^\circ}$
Hence, option **(1)** is the correct answer.

Q5) **Explanation:**

**Hint:** In an explosion, linear momentum is conserved

**Step 1:** Draw a diagram for the explosion.




**Step 2:** Find the speed of the $3$ kg.
Let's assume the identical parts are moving along the x and y axes of the coordinate system for simplicity, given that both parts are moving perpendicular to each other. By applying the conservation of linear momentum;
$0 = 1 \times 30\hat{i} + 1 \times 30\hat{j} + 3(\vec{v})$
$\Rightarrow \vec{v} = -(10\hat{i} + 10\hat{j})$
$\Rightarrow |\vec{v}| = 10\sqrt{2}$ m/s
Hence, option (1) is the correct answer.

Q6) **Explanation:**

**Hint:** $F_c = m\omega^2 r$

**Step:** Find the centripetal force on the car.
The magnitude of the centripetal force is given by; $F_c = m\omega^2 r$
$\Rightarrow F_c = 200 \times (0.2)^2 \times 70$
$\Rightarrow F_c = 8 \times 70$ N $= 560$ N
Therefore, the magnitude of centripetal force on the car is $560$ N.
Hence, option **(1)** is the correct answer.

Q7) **Explanation:**
**Hint:** The friction coefficient depends upon the nature of the contact surface.
**Step:** Recall friction.
The friction coefficient depends upon the surface's smooth or rough nature. It does not depend on normal force.

Q8) **Explanation:**

**Hint:** $p = \vec{F}.\vec{v}$

**Step:** Find the instantaneous power supplied to the particle.
The instantaneous power supplied to the particle is given by;
$p = \vec{F} \cdot \vec{v}$
$\Rightarrow p = (10\hat{i} + 10\hat{j} + 20\hat{k}) \cdot (5\hat{i} - 3\hat{j} + 6\hat{k})$
$\Rightarrow p = 50 - 30 + 120$
$\Rightarrow p = 140$ W
Hence, option **(4)** is the correct answer.

Q9) **Explanation:**

**Hint:** Find the relation between acceleration and time, using Newton's second law.

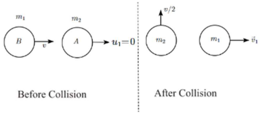
**Step 1:** Find the maximum frictional force acting on the object.
$f_{max} = \mu N$
$\Rightarrow f_{max} = 0.2 \times 3 \times 10 = 6$ N
$F \leqslant \mu N, a = 0$

**Step 2:** Find the acceleration of the particle.
$F > \mu N, a = \frac{F-f}{m}$
$\Rightarrow a = \frac{2t-6}{3}$
$\Rightarrow a = \frac{2}{3}t - 2$
Therefore, acceleration is zero until the applied force is less than the maximum frictional force and afterward increases linearly with time i.e.,
$F \leq \mu N, a = 0$
$F > \mu N, a = \frac{2}{3}t - 2$
Hence, option **(3)** is the correct answer.

Q10) **Explanation:**

**Hint:** The linear momentum of the system remains conserved.

**Step 1:** Find the velocity of the sphere $A$ after the collision.
The velocity of the particle before and after the collision is shown in the figure below;




Let the sphere of mass $A$ after collision move with velocity $\vec{v_1}$
By applying the conservation of linear momentum we get;
$\Rightarrow m_2 v\hat{i} + m_1 \times 0 = m_2(\frac{v}{2})\hat{j} + m_1\vec{v}_1\hat{i}$
By assuming, $m_1 = m_2 = m$
$\Rightarrow mv\hat{i} = m\left(\frac{v}{2}\right)\hat{j} + m\vec{v}_1$
$\Rightarrow \vec{v}_1 = v\hat{i} - \frac{v}{2}\hat{j}$

**Step 2:** Find the direction of the sphere of mass $A$ after collision.
The direction of the sphere is given by;
$\Rightarrow \tan\theta = \frac{v_y}{v_x} = \frac{\frac{-v}{2}}{v} = \frac{-1}{2}$
$\Rightarrow \theta = \tan^{-1}\left(\frac{-1}{2}\right)$
Therefore, the direction of the sphere of mass $A$ is $\theta = \tan^{-1}\left(\frac{-1}{2}\right)$ to the positive $x$-direction.
Hence, option **(4)** is the correct answer.

Q11) **Explanation:**

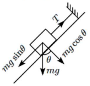
**Hint:** Reading of spring balance depends on the normal reaction.

**Explanation:** When the bird is sitting in the cage: The reading on the spring balance, $W_1$, includes the weight of the bird and the cage means the normal reaction is given by; $W_1 = mg_{\text{cage}} + mg_{\text{bird}}$
When the bird flies inside the cage: If the bird is simply hovering (in a non-flapping mode), then the normal reaction is given by;
$W_2 = mg_{\text{cage}}$
Therefore, the reading $W_1$ could be greater than $W_2$.
Hence, option **(2)** is the correct answer.

Q12 **Explanation:**

**Hint:** Balance the forces acting on the rope.

**Step:** Find the tension exerted by the rope to support the painting.
The free-body diagram of the system is shown in the figure below;




Since the surface is smooth and frictionless, there is no friction to resist the motion. The only force preventing the painting from sliding down the incline is the tension in the rope, which must balance the gravitational force parallel to the incline i.e., $T = mg\sin\theta$
Hence, option **(2)** is the correct answer.

Q13) **Explanation:**

**Hint:** $P = \frac{W}{t}$

**Step 1:** Find the initial power delivered.

$$\begin{aligned} W &= 25\ \text{kJ} \\ t &= 250\ \text{s} \\ P_i &= \frac{W}{t} \\ &= \frac{25\times10^3}{250} \\ &= 100\ \text{W} \end{aligned}$$

**Step 2:** Find the new power output of the motor.

$$\begin{aligned} P_f &= P_i + \frac{50}{100}\times P_i \\ &= 100 + \frac{50}{100}\times 100 \\ &= 150\ \text{W} \end{aligned}$$

Hence, option (4) is the correct answer.

Q14) **Explanation:**

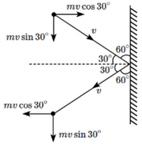
**Hint:** $\vec{p} = m_1\vec{v}_1 + m_2\vec{v}_2$

**Step:** Find the magnitude of the total momentum of the system before the collision.
As there is no external force applied to the two carts then the linear momentum of the system is conserved.
Given: $m_A = 1$ kg, $m_B = 2$ kg, $v_A = 0.2\hat{i}$ m/s, $v_B = -0.7\hat{i}$ m/s
The initial momentum of the system is given by;
$P_i = m_A v_A + m_B v_B = 1(0.2) - 2(0.7) = -1.2$ kg m/s
$\Rightarrow |P_i| = 1.2$ kg m/s
Hence, option (1) is the correct answer.

Q15) **Explanation:**

**Hint:** $F = \frac{\Delta p}{\Delta t}$

**Step 1:** Find the change of momentum of the ball by the wall.
The momentum of the ball before and after the collision is shown in the figure below;




The initial momentum of the ball is given by;
$p_i = mv\cos 30^\circ\hat{i} + mv\sin 30^\circ(-\hat{j})$
The final momentum of the ball is given by;
$p_f = mv\cos 30^\circ(-\hat{i}) + mv\sin 30^\circ(-\hat{j})$
By applying conservation of the linear momentum we get;
$p_i = p_f$
$\Rightarrow \Delta p = p_f - p_i = -2mv\cos 30^\circ\hat{i}$

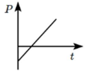
**Step 2:** Find the average force exerted on the ball by the wall.
The average force exerted on the ball is given by;
$\vec{F} = \frac{\Delta\vec{p}}{\Delta t}$
$\Rightarrow |\vec{F}| = \left|\frac{\Delta p}{\Delta t}\right| = \frac{2mv\cos 30^\circ}{0.20}$
$\Rightarrow |\vec{F}| = \frac{2\times 3\times 10\times\frac{\sqrt{3}}{2}}{0.20} = 150\sqrt{3}$ N
Hence, option **(4)** is the correct answer.

Q16) **Explanation:**

**Hint:** $P = \vec{F}.\vec{v}$

**Step 1:** Find the power delivered to the projectile.
The particle is projected at a speed $v_0$, then $\vec{v}_0 = v_x\hat{i} + v_y\hat{j}$
The force acting on the particle is given by; $F = -mg\hat{j}$
The power delivered to the projectile is given by;
$P = \vec{F} \cdot \vec{v}$
$\Rightarrow P = -mg\hat{j} \cdot (v_x\hat{i} + v_y\hat{j})$
$\Rightarrow P = -mgv_y$
For the projectile motion, the verticle velocity of the particle is given by;
$v_y = u_y - gt$
Then the power delivered to the projectile is given by;
$P = -mg(u_y - gt) = mg^2t - mgu_y$
$\Rightarrow P = mg^2t - mgu_y \ \ldots(1)$

**Step 2:** Find the correct graph for power $(P)$ vs time $(t)$.
The general equation of straight line is given by;
$y = mx - C \ \ldots(2)$
From equation $(1)$ and $(2)$ we get;
The power delivered is directly proportional to the time. Therefore, the graph should be a straight line with a negative $y$-intercept and a positive $x$-intercept.
Therefore, the correct graph for power $(P)$ vs time $(t)$ is given below;




Hence, option **(4)** is the correct answer.

Q17) **Explanation:**

**Hint:** Explosion occurs due to internal forces.

**Explanation:**
The path of fragments is such that the centre of mass of the fragments moves in the same path, which the firecracker would have followed, had it not exploded. In reality, when a firecracker explodes, the fragments move in different directions due to the internal forces exerted during the explosion. The individual fragments do not continue in the same exact path as the original firecracker. Therefore, Assertion (A) is false because it incorrectly implies that all fragments individually follow the same path, rather than describing the path of the center of mass.
The explosion occurs due to internal forces only. The explosion involves internal forces (from chemical reactions within the firecracker), and no external forces (neglecting air resistance) act on the firecracker system during the explosion. This is why the center of mass of the system continues on the same path, even though individual fragments do not.
Hence, option **(4)** is the correct answer.

Q18) **Explanation:**

**Hint:** Work done = $\Delta$U

**Step 1:** Find an increase in the potential energy of a brick.
$\Delta U$ = mgh

**Step 2:** Find the energy supplied by man in 1 h.
E = P(3600) J.

**Step 3:** Find the number of bricks.
$E = n(\Delta U)$
$n = \dfrac{9.8 \times 3600}{2.5 \times 9.8 \times 3.6}$
$= 400$

Q19) **Explanation:**

**Hint:** The change in kinetic energy of a particle is equal to the work done on it by the net force.

**Explanation:** The Work-Energy Theorem states that the net work done by all forces on a body equals the change in its kinetic energy. This is a fundamental principle in mechanics.
Hence, **Statement I** is correct.
Non-conservative forces (e.g., friction, air resistance) can perform work, and this work often transforms mechanical energy (kinetic or potential) into other forms of energy (e.g., heat). However, this does not mean that non-conservative forces cannot change the kinetic energy of a system. In fact, they can affect the kinetic energy.
Hence, **Statement II** is incorrect.
The work-energy theorem states that the change in kinetic energy of a particle is equal to the work done on it by the net force. The work done by all forces changes the kinetic energy of the system.
Hence, option **(4)** is the correct answer.

Q20) **Explanation:**

**Hint:** $a = \mu g$

**Step 1:** Find the acceleration of the block.
Given: $u = 20$ m/s, $s = 40$ m, $v = 0$ m/s
Using the third equation of motion we get;
$v^2 - u^2 = 2as$
$\Rightarrow 0^2 - (20)^2 = 2 \times a \times 40$
$\Rightarrow a = -\frac{20\times20}{40\times2} = -5$ m/s$^2$

**Step 2:** Find the coefficient of kinetic friction.
The friction force acting on the body is given by;
$f_k = \mu_k N$
$\Rightarrow f_k = \mu_k mg$
$\Rightarrow \frac{-f_k}{m} = -5 \quad [f_k = -ve]$
$\Rightarrow -\frac{\mu_k mg}{m} = -5$
$\Rightarrow \mu_k \times 10 = 5$
$\Rightarrow \mu_k = 0.5$
Therefore, the coefficient of kinetic friction is $0.5$.
Hence, option **(4)** is the correct answer.

Q21) **Explanation:**

**Hint:** Apply the conservation of energy.

**Step:** Find the increase in the kinetic energy.
As the block $B$ falls through a height $h$, the potential energy decreases by $mgh$. The decrease in potential energy is used to increase the kinetic energy of both blocks. Therefore the increase in kinetic energy of each block is $\frac{mgh}{2}$.
Hence, option (3) is the correct answer.

Q22) b is the correct option

Q23) **Explanation:**

**Hint:** Apply the work-energy theorem.

**Explanation:** Both the ground frame and the lift frame are inertial. The lift is moving at a constant velocity. The change in kinetic energy is zero in both frames of reference. According to the work energy theorem, the work done by all forces is equal to the change in kinetic energy of the particle.
$W = \Delta K.E = 0$
$\Rightarrow W_N + W_g = 0$
Hence, option (4) is the correct answer.

Q24) **Explanation:**

**Hint:** Recall Work-energy theorem.

**Step:** Analyse each statement one by one.
**(A)** Every force encountered in mechanics has an associated potential energy. This statement is incorrect because only conservative forces have an associated potential energy. Non-conservative forces, such as friction, do not have a potential energy function associated with them.
**(B)** The work done by all the forces is equal to the change in kinetic energy. This statement is correct because this describes the work-energy theorem, which states that the net work done by all the forces on an object is equal to the change in its kinetic energy.
**(C)** The work done by all the forces is equal to the change in potential energy. This statement is incorrect because the work done by conservative forces is related to the change in potential energy, but the total work done by all forces (including non-conservative forces) relates to the change in kinetic energy, not potential energy.
**(D)** The total mechanical energy of a system is conserved if the forces, doing work on it, are conservative. This statement is correct because if only conservative forces do work on a system, the total mechanical energy (the sum of kinetic and potential energy) remains constant.
Therefore, the correct statements are **(B)** and **(D)**.
Hence, option **(4)** is the correct answer.

Q25) **Explanation:**

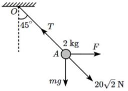
**Hint:** $Mg - N = Ma$

**Step:** Find the acceleration of the lift.
The lift is accelerating downward with acceleration $a$ then the force equation is given by;
$Mg - N = Ma$
$\Rightarrow Mg - \frac{Mg}{4} = Ma \qquad \left[N = \frac{Mg}{4}\right]$
$\Rightarrow \frac{3Mg}{4} = Ma$
$\Rightarrow a = \frac{3g}{4}$
Hence, option **(3)** is the correct answer.

Q26) **Explanation:**

**Hint:** Balance the horizontal and vertical forces.

**Step:** Find the value of tension in the string.
The force acting on the bob $(A)$ is shown in the figure below;

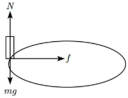



The horizontal force acting on the bob is $F$ N and the vertical force acting on the bob is $mg = 20$ N
The string makes an angle of $45^\circ$ with the vertical & the system is stationary then,
The resultant of these two forces is balanced by tension in the string i.e., $T = 20\sqrt{2}$ N
Hence, option **(4)** is the correct answer.

Q27) **Explanation:**

**Hint:** $v_{max} = \sqrt{\mu g r}$

**Step 1:** Draw the free-body diagram of the cyclist and find the friction force.
The free-body diagram of the cyclist is shown in the figure below;



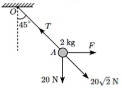

As the cyclist is not tilting so, the frictional force is providing the necessary centripetal force.
The frictional force acting on the cyclist is given by; $f = \mu N = \mu mg$

**Step 2:** Find the minimum value of the coefficient of static friction.
Given: $v = 5$ m/s, $r = 5$ m
For the cyclist to take a circular turn, $f \geq \frac{mv^2}{r}$
$\Rightarrow \mu mg \geq \frac{mv^2}{r}$
$\Rightarrow \mu \geq \frac{v^2}{rg}$
$\Rightarrow \mu \geq \frac{(5)^2}{5\times10} \Rightarrow \mu \geq \frac{1}{2}$
The minimum value of the coefficient of static friction is $0.50$.
Hence, option (1) is the correct answer.

Q28) **Explanation:**

**Hint:** Balance the horizontal and vertical forces.

**Step:** Find the value of the force $F$.
The force acting on the bob ($A$) is shown in the figure below;

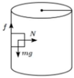



The horizontal force acting on the bob is $F$ N and the vertical force acting on the bob is $mg = 20$ N
The resultant of these two forces is balanced by tension in the string.
The string makes an angle of $45^\circ$ with the vertical & the system is stationary then, horizontal and vertical force should be equal i.e., $F = 20$ N
Hence, option **(2)** is the correct answer.

Q29) **Explanation:**

**Hint:** The friction force balances the weight of the block.

**Step 1:** Find the normal reaction of the block.




The net force on the body towards the centre is $F_c = mr\omega^2$
The normal reaction of the block is given by;
$N = mr\omega^2$

**Step 2:** Find the minimum angular velocity.
The frictional force is given by;
$f = \mu N$
Now, the block is in rest in the vertical direction.
$f = \mu N$
Subsitute the known values we get,
$mg = \mu mr\omega^2$
$\Rightarrow \omega = \sqrt{\frac{g}{\mu r}}$
$\Rightarrow \omega = \sqrt{\frac{10}{0.1 \times 1}}$
$\Rightarrow \omega = 10$ rad/s
Hence, option **(4)** is the correct answer.

Q30) **Explanation:**

**Hint:** $F_{net} = ma$

**Explanation:** The force acting on the train is constant, and since the train's mass does not change, the acceleration remains constant according to Newton's Second Law $F_{net} = ma$.
Acceleration, which is the rate of change of velocity, ensures that as long as it remains constant and acts in the direction of motion, the train's velocity will continue to increase over time.
Therefore, the acceleration of the train remains constant, while its velocity steadily increases.
Hence, option **(2)** is the correct answer.

Q31) **Explanation:**

**Hint:** $W = \vec{F}.\vec{s}$

**Explanation:** The work done on the man by the reaction force exerted by the stairs is zero because the reaction force applied by the staircase is normal to the man but there is no displacement due to this force.
Hence, option (4) is the correct answer.

Q32) **Explanation:**

**Hint:** Work done=Force×displacement

**Step:** Find the work done by road on the cycle.
The work done by road on the cycle is given by;
$W = \vec{F} \cdot \vec{s} = Fs\cos\theta$
$\Rightarrow W = 200 \times 10\cos 180^\circ = -2000$ J
Hence, option **(3)** is the correct answer.

Q33) **Explanation:**

**Hint:** The astronaut is in free fall initially.

**Step 1:** Find the net initial acceleration of the astronaut.
As the astronaut is in free fall, the initial apparent weight of the astronaut is equal to zero.

**Step 2:** Find the final apparent weight of the astronaut.
When he accelerates with an acceleration equal to $\frac{3g}{4}$ in the forward direction,
his apparent weight = $\frac{3mg}{4}$ = 75% of his actual weight = 25% less than his actual weight
Hence, option **(2)** is the correct answer.

Q34) **Explanation:**

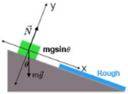
**Hint:** $f_{max} = \mu_s N$

**Step 1:** Find the coefficient of static friction.
When the block starts to move,
$F = f_{max} = \mu_s N$
$\mu_s = \frac{20}{50} = 0.4$

The coefficient of kinetic friction will be less than the coefficient of static friction.

Q35) **Explanation:**

**Hint:** The change in kinetic energy of the block is zero.




**Step 1:**

If s is the length along the incline
**Force approach :**
**Step 1 :** Find the final velocity for the first half.
For top half,
$a = g\sin\theta$
$v^2 = 0 + 2a\left(\frac{s}{2}\right) \quad .......(1)$
**Step 2 :** Find the final velocity for the second half.
For bottom half,
$a = -(\mu g\cos\theta - g\sin\theta)$
$0 = v^2 + 2a\left(\frac{s}{2}\right) \quad .......(2)$
Equation (1) + (2) :
$\mu g\cos\theta = 2g\sin\theta$
$\mu = 2\tan\theta$
**Energy approach :**
**Step 1 :** Apply work − energy theorem.
Gain in kinetic energy = loss of kinetic energy due to friction
$mg(s \times \sin\theta) = (\mu mg\cos\theta)\frac{s}{2}$
$\mu = 2\tan\theta$

Q36) **Explanation:**

**Hint:** Use the work-energy theorem.

**Step:** Find the work done by the gravitational force and the resistive force of air.
Using the work-energy theorem we get;
$W_g + W_a = \Delta KE$
$mgh + W_a = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$
Subsitute the known values we get;
$W_a = \frac{1}{2}m(50)^2 - \frac{1}{2}m(0)^2 - m \times 10 \times 1000$
$W_a = \frac{1}{2} \times 0.001 \times 2500 - 0.001 \times 10000$
$W_a = \frac{2.5}{2} - 10$
$\Rightarrow W_a = -8.75\ \mathrm{J}$ and,
$W_g = mgh$
$= 0.001 \times 10 \times 1000$
$\Rightarrow W_g = 10\ \mathrm{J}$
Hence, option **(3)** is the correct answer.

Q37) **Explanation:**

**Hint:** $W = F \cdot d$

**Step:** Find the work done in displacing the particle.
Given: $\vec{F} = 2\hat{i}$ N, $\vec{r} = (3\hat{i} + 4\hat{j})$ m
The work done on the particle is given by;
$W = \vec{F} \cdot \vec{r} = (2\hat{i}) \cdot (3\hat{i} + 4\hat{j}) = 6$ J
Hence, option (3) is the correct answer.

Q38) option b is correct

Q39) **Explanation:**

**Hint:** $r = \sqrt{x^2 + y^2}$
**Step 1:** Find the acceleration of the block.


$a_y = \dfrac{5}{5} = 1\ \mathrm{m/s^2}$
$a_x = 0$
**Step 2:** Find the displacement in the x & y-direction.
$S_x = u_x t$
$= 1.5 \times 4 = 6\ \mathrm{m}$
$S_y = 0 + \dfrac{1}{2}(1)(4)^2$
$= 8m$
**Step 3:** Find the net displacement.
$r = \sqrt{s_x^2 + s_y^2}$
$= 10\ \mathrm{m}$
Hence, option **(4)** is the correct answer.

Q40) **Explanation:**

**Hint:** $\Delta W = K.E_f - K.E_i$

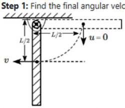
**Explanation:** According to the work-energy theorem: The change in kinetic energy of a particle is equal to the work done on it by the net force (all forces, conservative or non-conservative). Therefore, **(A)** is false.
According to Newton's laws of motion, the acceleration of any particle of the system is due to all forces acting on it; and therefore the work done on it by these is the total work done. Therefore, **(R)** is true.
Therefore, **(A)** is false but **(R)** is true.
Hence, option (4) is the correct answer.

Q41) option a is correct

Q42) **Explanation:**

**Hint:** Conserve the mechanical energy of the rod.

**Step 1:** Find the final angular velocity of the rod.




Conserve the mechanical energy of the rod we get;

$\Rightarrow U_i + K_i = U_f + K_f$

$\Rightarrow Mg\frac{L}{2} + 0 = 0 + \frac{1}{2}I\omega^2 = \frac{1}{2}\frac{ML^2}{3}\omega^2$

$\Rightarrow \omega^2 = \frac{3g}{L}$

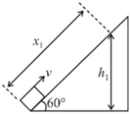
**Step 2:** Find the force by the hinge on the rod.
Using Newton's second law of motion, we get;

$\Rightarrow F - Mg = \frac{ML\omega^2}{2}$

$\Rightarrow F = \frac{3Mg}{2} + Mg = \frac{5}{2}Mg = 2.5Mg$

Hence, option **(4)** is the correct answer.

Q43) Case 1:

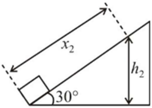



$x_1 \sin 60^\circ = h_1$

Using energy approach-

$\frac{1}{2}mv^2 = mg\left(x_1 \sin 60^\circ\right) \qquad \ldots(1)$

Case 2:




$x_2 \sin 30^\circ = h_2$

$\frac{1}{2}mv^2 = mgh_2 = mg(x_2 \sin 30^\circ) \qquad \ldots(2)$

$From\ (1)\ and\ (2)$

$\frac{x_1}{x_2} = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$

Hence, option (4) is the correct answer.

Q44) **Explanation:**

**Hint:** The net force acting on an object is zero.

**Explanation:** According to Newton's second law of motion, the net force acting on the particle is the rate of change of momentum. Since the particle is moving with constant velocity, the change in momentum is zero and net force should be zero. Therefore, the two forces have equal magnitude but point in opposite directions.
Hence, option **(3)** is the correct answer.

Q45) **Hint:** When the velocity is constant, the acceleration is equal to zero.

**Explanation:** As blocks are moving at a constant speed, means all the blocks are moving with zero acceleration. Therefore, the net force acting on each block is also zero.
Hence, option (4) is the correct answer.

Q46) **Explanation:**

Hint: If NaOH is in excess then the solution will be basic, so $[OH^-]$ can be calculated.

$NaOH \quad is \quad in \quad excess. \quad Hence, \quad it \quad will \quad remain \quad in \quad the \quad solution$

$[OH \quad ]^- = \frac{total \ volume \ moles \ of \ NaOH \ -moles \ of \ HCl}{total \ vol}$

$Substitute \quad values \quad in \quad the \quad above \quad expression.$

$[OH]^- = \frac{50\times0.2-50\times0.1}{50+50} = 0.05$

$pOH = -\log[OH \quad ]^- = -\log 0.05 = 1.3$

$pH = 14 - pOH = 14 - 1.3 = 12.7$

Q47) Hint: Equilibrium constant for the reverse reaction is the inverse of the equilibrium constant for the reaction in the forward direction.

The given equation is as follows:

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ $\quad k = 4 \times 10^{-4}$

Convert the equation and multiple by 1/2.

$NO(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{1}{2}O_2(g)$ $k''$

The relation between k and $k''$ is as follows:

$k'' = (\frac{1}{k})^{\frac{1}{2}}$

$k'' = (\frac{1}{4\times10^{-4}})^{\frac{1}{2}}$

$k'' = 50$

Q48) **Explanation:**

Step 1: The equilibrium is represented as follows: $Cr(OH)_3 \rightleftharpoons \underset{s}{Cr^{3+}} + \underset{3s}{3OH^-}$

Let "s" be the solubility. So, the solubility product is written as follows:

$K_{sp} = s \times (3s)^3 = 27 s^4$

or, $6 \times 10^{-31} = 27 s^4$

or, $s = (\frac{6\times10^{-31}}{27})^{1/4}$

Step 2: The solubility of hydroxide ion is $3s$

$[OH^-] = 3s = 3 \times (\frac{6\times10^{-31}}{27})^{1/4}$

or, $[OH^-] = (18 \times 10^{-31})^{1/4}$

Thus, **option 2** is the correct choice.

Q49) **Hint:** The equilibrium constant value only depends on the temperature.

**Explanation:**

**(i)** The equilibrium constant in terms of pressure (Kp) is defined as the ratio of the partial pressures of the products to the partial pressures of the reactants at equilibrium, each raised to the power of their respective stoichiometric coefficients.

When the total pressure of a system is increased, the partial pressures of all gases in the system increase proportionally, assuming ideal gas behavior. However, since Kp is a ratio of these partial pressures, a uniform increase in pressure does not change the value of Kp.

**(ii)** The equilibrium constants Kp and Kc depend only on the temperature of the system and not on the degree of dissociation or pressure of the system.

Thus, **option 3** is the correct answer.

Q50) The given reaction can be converted as

$$\frac{1}{2}B + \frac{1}{2}C \rightleftharpoons A \quad \Rightarrow \quad K_1' = 1$$

$$\frac{1}{2}C + \frac{1}{2}D \rightleftharpoons B \quad \Rightarrow \quad K_2' = \frac{1}{4}$$

$$P \rightleftharpoons C + \frac{1}{2}D \quad \Rightarrow \quad K_3' = \frac{1}{5}$$

$$P \rightleftharpoons A + \frac{1}{2}B \quad \Rightarrow \quad K_4 = 1 \times \frac{1}{4} \times \frac{1}{5} = \frac{1}{20}$$

Q51) **Explanation:**

**Hint:** The solution is neutral when $K_a$ and $K_b$ are equal.

Since **Ka and Kb are equal**, the acidic strength of ammonium ions ($NH_4^+$) and the basic strength of acetate ions ($CH_3COO^-$) balance each other out. Therefore, the solution is **neutral**.

Q52) **Explanation:**

Step 1: $AB(g) + CD(g) \rightleftharpoons AD(g) + CB(g)$

The number of moles of $AB \text{ and } CD$ = 1

So, the concentration of $[AB] = [CD] = \frac{1}{V} \text{ mol L}^{-1}$

where V is the volume of the vessel in L.

The equilibrium constant, $K_c$, is the ratio of the equilibrium concentrations of products over the equilibrium concentrations of reactants, each raised to the power of their stoichiometric coefficients.

$$K_c = \frac{[AD][CB]}{[AB][CD]}$$

Step 2: It is given that, at equilibrium, $3/4$ mole of each reactant is converted to a product.

At equilibrium,

$$[AB] = [CD] = \frac{1-3/4}{V} = \frac{1}{4V} \text{ mol}^{-1}$$

$$[AD] = [CB] = \frac{3}{4V} \text{ mol}^{-1}$$

$$K_c = \frac{\left(\frac{3}{4V}\right)\left(\frac{3}{4V}\right)}{\left(\frac{1}{4V}\right)\left(\frac{1}{4V}\right)} = 9$$

So, **option 4** is the correct answer.

Q53) **Explanation:**

**Hint:** $K_a$ $\alpha$ Degree of dissociation

**Explanation:**

The dissociation constant is defined as the ratio of dissociated ions (products) to the original acid or bases in the solution(reactants). For acids, the acid dissociation constant is defined as the ratio of concentration of dissociated ions to the concentration of original acid.

The greater the dissociation constant of an acid the stronger the acid, more will be dissociation.

From the given molecules, HNCO (cyanic acid) has highest value of $K_a$ i.e. $3.3 \times 10^{-4}$

Thus, it has greatest extent of dissociation. Hence, option 2 is the correct answer.

Q54) **Explanation:**

**Hint:** $\mathrm{pH} + \mathrm{pOH} = 14$ (for any aqueous solution at $25^{\circ}\mathrm{C}$ )

**Explanation:**

Step 1: The pH of given solution is $\mathbf{4.7}$

We know that, for any aqueous solution at $\mathbf{25^{\circ}}$ ; $\mathbf{pH + pOH = 14}$

Thus, the pOH of the solution will be: $\boldsymbol{pOH = 14 - 4.7 = 9.3}$

Step 2: The pOH of any solution is equal to the negative logarithm of the hydroxide ion concentration.

i.e., $\boldsymbol{pOH = -log[OH^-]}$

or, $\boldsymbol{9.3 = -log[OH^-]}$

or, $\boldsymbol{[OH^-] = 10^{-9.3} = 5.01 \times 10^{-10}\ M}$

Thus, **option 2** is the correct answer.

Q55) **Explanation:**

**Hint:** Reaction shifts in the forward direction.

**Explanation:**

Endothermic reactions are favourable at high temperatures. When the temperature of an endothermic reaction is increased, the reaction will proceed in the forward direction so as to absorb heat and nullify the effect of increasing temperature.

This will increase the value of the equilibrium constant $K_c$ and increase the yield of the reaction. Hence, **option 2** is the correct answer.

Q56) **Explanation:**

$$Ag_2C_2O_4\ (s) \leftrightharpoons 2Ag^+(aq) + C_2O_4^{2-}(aq)$$
$$K_{SP} = [Ag^+]^2\,[C_2O_4^{2-}]$$
$$[Ag^+] = 2.2 \times 10^{-4}\ \mathrm{M}$$

Given that:
Concentration of $C_2O_4^{2-}$ ions,

$[C_2O_4^{2-}] = 2.2\times10^{-4} / 2 = 1.1\times10^{-4}$ M

Therefore $K_{sp} = (2.2\times10^{-4})^2(1.1\times10^{-4})$

$= 5.324 \times 10^{-12}$

Q57) **Explanation:**

Hint: Brönsted acids - Donate proton
Brönsted bases - Accept proton

Q58) Hint: The value of equilibrium constant is very large. Then reaction almost completed

Step 1:

If $K_c > 10^3$, products predominate over reactants, i.e., if $K_c$ is very large, the reaction proceeds nearly to completion. The concentration of the product is very high as compared to the reactant.

Step 2:

$$A \rightleftharpoons B$$

$$K_c = \frac{[B]}{[A]} = 1.6 \text{ X } 10^{12}$$

The equilibrium constant value is very high, thus, the concentration of the product is more as compared to the reactant.

Q59) Hint: Conjugate base of strong acid and conjugate acid of a strong acid is stable in the solution hence does not undergo a hydrolysis reaction

A salt of a strong acid with a strong base does not undergo hydrolysis. In this case, neither the cations nor the anions undergoes hydrolysis. Therefore the solution remains neutral.
The cation and anion generated by salt are stable and hence do not reacts with water.

These salts undergo hydration. Here water is used as a solvent. Water molecules surround and interact with cations and anions.

Hence, option second is the correct answer.

Q60) **Explanation:**

**Hint:** $K_P = P_{NH_3}^2 \times P_{CO_2}$

$$H_2NCOONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$$

2x=12 atm x = 6 atm

$$K_P = P_{NH_3}^2 \times P_{CO_2}$$
$$= (12)^2 \times 6$$
$$= 144 \times 6$$
$$= 864 \text{ atm}^3$$

Q61) Hint: Use the formula of dilution that is, $M_1 \times V_1 = M_2 \times V_2$

Step 1:

Calculate the concentration of $H^+$ ion when pH is 1 and pH is two respectively.

pH = - log[$H^+$]

1 = - log[$H^+$]

[$H^+$] = $10^{-1}$

Same ar pH 2, [$H^+$] = $10^{-2}$ . $H^+$ ion concentration tell about the concentration of acid.

Step 2:

Calculate the amount of water in litres must be added to 1 litre of an aqueous solution of HCl as follows:

$$M_1 \times V_1 = M_2 \times V_2$$
$$10^{-1} \times (1L) = 10^{-2} \times V_2$$
$$V_2 = 10 \text{ L.}$$
$$V_{water} = V_2 - V_1 = 9 \text{ L}$$

Hence, option second is the correct answer.

Q62) **Explanation:**

Hint: $\mathbf{pH} = 7 + \frac{1}{2}(\mathbf{pK_a} - \mathbf{pK_b})$

Calculate the pH of dimethylammonium acetate is as follows:

$$\mathbf{pH} = 7 + \frac{1}{2}(\mathbf{4.77} - \mathbf{3.27})$$
$$\mathbf{pH} = \mathbf{7.75}$$

Hence, option 1 is the correct answer.

Q63) The reaction is as follows:

$Mg(OH)_2 \Leftrightarrow Mg^{2+} + 2OH^-$

The $K_{sp}$ formula for $Mg(OH)_2$ is as follows:

$K_{sp} = [Mg^{2+}][OH^-]^2$

The solubility is S.
$K_{sp} = [S][2S]^2$
$= 4S^3$

Q64) **Explanation:**

Hint: Acidic strength is directly proportional to the oxidation state of the central atom
The acidic strength of oxyacids increases with an increase in electronegativity of the non-metal and its oxidation number.

The oxidation state of the central atom in a compound is as follows:

$$\mathbf{HCIO_4} = 7 \quad \mathbf{H_2SO_4} = +6 \quad \mathbf{H_3PO_4} = +5$$

Thus the order is

$$\mathbf{HCIO_4(III)} > \mathbf{H_2\,SO_4(I)} > \mathbf{H_3\,PO_4(II)}$$

Q65) **Explanation:**

**Hint:** Density and molecular mass of a pure substance is always fixed at particular temperature.
**Explanation:**
For a pure substance (both solids and liquids),

$$\begin{aligned}[\text{Pure}]\ \text{substance} &= \frac{\textit{Number}}{} \ of\ molesVolume \\ &= \frac{\textit{Mass/molecular}}{}\ massVolume \\ &= \frac{\textit{Mass}}{\textit{Volume} \times \textit{Molecular}}\ mass \\ &= \frac{\textit{Density}}{\textit{Molecular}}\ mass\end{aligned}$$

Now, the molecular mass and density (at a particular temperature) of a pure substance is always fixed and is accounted for in the equilibrium constant.

Therefore, the values of pure substances are not mentioned in the equilibrium constant expression.

Q66) **Explanation:**

Step 1: The solubility product ($K_{sp}$)of $PbCl_2$ = $3.2 \times 10^{-8}$

The equilibria for $PbCl_2$ is represented as: $PbCl_2 \rightleftarrows \underset{s}{Pb^{2+}} + \underset{2s}{2Cl^-}$

where "s" is the solubility in "mol $L^{-1}$"

Thus, the solubility product is written as: $K_{sp} = s \times (2s)^2 = 4s^3$

$3.2 \times 10^{-8} = 4s^3$

or, $s = 2 \times 10^{-3}\ mol\ L^{-1}$

Step 2: The molar mass of $PbCl_2$ = $278\ g\ mol^{-1}$

The given mass is 0.1 g

By using the formula of molarity, the volume of water can be calculated:

$$2 \times 10^{-3}\ mol\ L^{-1} = \frac{0.1\ g}{278\ g\ mol^{-1} \times V}$$

$$V = \frac{0.1\ g}{278\ g\ mol^{-1} \times 2 \times 10^{-3}\ mol\ L^{-1}} = 0.18L$$

Hence, **option 4** is the correct choice.

Q67) Explanation:

**Hint:** The addition of inert gas at constant volume will not affect the partial pressure of the reactant or products.

**Explanation:**

The given reaction is: $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

The addition of an inert gas of constant volume condition to an equilibrium has no effect hence reaction remains unaffected. This is because the addition of inert gases at constant volume increases the total pressure. However, it does not have partial pressures of the reactants and products.

The mole fraction remains the same. Because the reactants and products are unaffected in terms of their moles, the mole fractions and thus their partial pressures remain constant even though the overall pressure rises.

Hence, **option 1** is the correct answer.

Q68) Explanation:

**Hint:** $\mathrm{p}K_\mathrm{a} = -\log K_\mathrm{a}$

**Explanation:**

The $pK_a$ value of an acid is a measure of its acidity, with lower $pK_a$ values indicating stronger acids. The relationship between $pK_a$ and acid strength is logarithmic, as the $pK_a$ is derived from the equilibrium constant for the dissociation of the acid: $\mathrm{p}K_\mathrm{a} = -\log K_\mathrm{a}$

$K_a$ of $A = 10^{-4}$
$K_a$ of $B = 10^{-5}$

i.e., A is 10 times stronger than B. Since Acid A has a lower $pK_a$ value, it is stronger than Acid B.

Hence, **option 1** is the correct answer.

Q69) option a is correct

Q70) Hint: Salt having a low $\mathbf{K_{sp}}$ value will precipitate first
Let solubility be S

$$\mathbf{Ag_2\,CrO_4 \rightarrow 2\,Ag^{+}\,2S + \underset{S}{CrO_4^{2-}}}$$

$$\mathbf{K_{sp} = 4s^3}$$
$$\mathbf{s = \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{10^{-12} \times \left(\frac{1.1}{4}\right)} = 7.2 \times 10^{-4}}$$

$$\mathbf{AgBr = \sqrt{5 \times 10^{-13}} = 7.07 \times 10^{-6}}$$

$$\mathbf{AgCl = \sqrt{1.8 \times 10^{-10}} = 1.34 \times 10^{-5}}$$

$$\mathbf{AgI = \sqrt{8.3 \times 10^{-17}} = 9.11 \times 10^{-9}}$$

Hence, the solubility of $\mathbf{Ag_2\,CrO_4}$ is high so, it will precipitate at the last.

Q71) option a is correct

Q72) Hint: Use the formula, $K_p = K_c\,(RT)^{\Delta n_g}$

Step 1:

The reaction is as follows:

$Fe_2N_{(s)} + \frac{3}{2}H_{2(g)} \leftrightharpoons 2Fe_{(s)} + NH_{3(g)}$

The relation between $K_p$ and $K_c$ are as follows:

$K_p = K_c\,(RT)^{\Delta n_g}$

Step 2:

Calculate the value of $\Delta n_g$ as follows:

$\Delta n_g = 1 - \frac{3}{2}$

$= -\frac{1}{2}$

The relation is as follows:

$K_p = K_c\,(RT)^{\frac{-1}{2}}$

$K_c = K_p(RT)^{1/2}$

Q73) **Explanation:**

**Hint:** Common ion effect.

Q74) option d is correct

Q75) **Explanation:**

**Hint:** A catalyst does not affect the final state of equilibrium.

**Explanation:**

Catalysts are any substance that increases the rate of a reaction without itself being consumed. Enzymes are naturally occurring catalysts responsible for many essential biochemical reactions.

A catalyst provides an alternative reaction pathway with a lower activation energy, allowing the reaction to proceed faster in both the forward and reverse directions, but it does not alter the final equilibrium position. The primary effect of a catalyst is to increase the rate of both the forward and reverse reactions equally, leading to a faster attainment of equilibrium.

it describes the catalyst's mechanism) but misleading regarding equilibrium.
Both assertion and reason are false. Hence, **option 4** is the correct answer.

Q76) Hint: The acid-base pair that differs only by one proton is called a conjugate acid-base pair.

Step 1:

The given reaction is an example of an acid-base reaction. The reaction is as follows:
$CH_3O^- + H_2O \rightarrow CH_3OH + OH^-$

$CH_3O^-$ acts as a base and $H_2O$ acts as an acid. $CH_3OH$ is a conjugate acid of $CH_3O^-$ and $OH^-$ is a conjugate base of $H_2O$.

Step 2:

The first statement is a correct statement.

The second statement is the wrong statement because $CH_3O^-$ ion is a Lewis base, not a Lewis acid.
The third statement is also a wrong statement. The pH of a final mixture of solution value is more than 7 because the final mixture is basic in nature. In the final mixture $OH^-$ is formed.

The fourth statement is also wrong. Water is acting as a Bronsted–Lowry acid, not a base because it gives $H^+$ ion.

Hence, an option first is the correct answer.

Q77) Hint: $K_P = P_{Hg} \times P_{O_2}^{\frac{1}{2}}$

Step 1:

$HgO(s) \rightleftharpoons Hg(g) + \frac{1}{2}O_2(g)$

1 0 0

t =t , (1-x) x x/2

Solid substance mole remained constant and did not include in equilibrium constant expression.

Total moles at equilibrium = $x + \frac{x}{2}$

$= \frac{3x}{2}$

Step 2:

Calculate the partial pressure of gaseous products and reactants using dalton's law of partial pressure

$P = P_T X(\text{mole fraction})$

$P_{Hg} = \frac{x}{3x/2} P$

$P_{Hg} = \frac{2}{3}P$

$P_{O_2} = \frac{x/2}{3x/2} P = \frac{1}{3}P$

$K_P = P_{Hg} \times P_{O_2}^{\frac{1}{2}}$

$K_p = \frac{2}{3}P\left(\frac{1}{3}P\right)^{1/2} = \frac{2}{3^{3/2}} P^{3/2}$

Hence, option 1 is the correct answer.

Q78) For $A(s) \rightleftharpoons 2B(g) + 3C(g)$.

$\therefore$ $K_c = [C]^3[B]^2$; if [C] becomes twice,

Then let conc. of B becomes B' then

$K_c = [2C]^3[B']^2$

or $[C]^3 . [B]^2 = [2C]^3[B']^2$

$\therefore$ $\frac{[B']}{[B]} = \frac{1}{8} = \frac{1}{2\sqrt{2}}$

Q79) **Hint:** If $Q_p > K_p$, the reaction will proceed in the backward direction.

**Explanation:**

**Step 1:** Calculate $Q_p$

For the reaction:

$$FeO_{(s)} + CO_{(g)} \rightleftharpoons Fe_{(s)} + CO_{2(g)}$$

Initial p 1.4 atm

$Q_p = \frac{p_{co_2}}{p_{co}} = \frac{0.80}{1.4}$

$Q_p = 0.571$ and $K_p = 0.265(\text{given})$

If $Q_p > K_p$, the reaction will proceed in the backward direction.

**Step 2:** Find partial pressure of $CO_2$

Therefore, we can say that the pressure of CO will increase while the pressure of $CO_2$ will decrease.

Now, let the increase in pressure of CO = decrease in pressure of $CO_2$ be p.

Then, we can write,

$K_p = \frac{p_{CO_2}}{p_{CO}}$

$\Rightarrow 0.265 = \frac{0.80 - p}{1.4 + p}$

$\Rightarrow p = 0.339$ atm

Therefore , equilibrium partial pressure of $CO_2$, $p_{CO_2}$= 0.80 - 0.339 = 0.46 atm

Q80) **Explanation:**

**Hint:** Solid reactants do not affect the concentration.

**Explanation:**

Le Chatelier's principle states that if a dynamic equilibrium is disturbed by changing the conditions, the position of equilibrium shifts to counteract the change to reestablish equilibrium.

However, Le Chatelier's does not apply to pure solids and liquids. We can't increase the concentration of a pure solid or liquid because they are the most concentrated form of the substance. Their effective concentration is constant throughout the reaction.

So, the addition of $Fe_2O_3$ will not affect the equilibrium as it is a solid reactant. Other reactants are in gaseous form and will affect the reaction's equilibrium. Thus, **option 4** is the correct answer.

Q81) **Explanation:**

**Hint:** Salt of strong base and weak acid are basic in nature.

**Explanation:**

The solution of sodium acetate is basic in nature.

The reaction is as follows:

$$\mathbf{CH_3\,COONa \Rightarrow Salt\ of\ CH_3\,COOH(WA) + NaOH(SB)}$$
$$\mathbf{\therefore\ Solution\ of\ CH_3\,COONa\ shows\ basic\ nature,.}$$

Q82) Hint: Use Henderson–Hasselbalch equation

Step 1:

The Henderson–Hasselbalch equation is as follows:

$$pH = pK_a + \log \frac{[\text{Salt}]}{[\text{ Acid }]}$$

The given values are as follows:

$pK_a = 4.57$
[Salt] = 0.10 M
[Acid] = 0.01 M

Step 2:

Calculate the pH as follows:

$$pH = 4.57 + \log \frac{[0.10]}{[0.01]}$$
= 4.57 + log10
= 4.57 + 1
pH = 5.57

Q83) **Explanation:**

**Hint:** Use formula $K_C = \frac{K_p}{(RT')^{\Delta n}}$

**Explanation:**

The given reaction is as follows: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

where, $\Delta n$ = Number of gaseous products - Number of gaseous reactants

It is given that, $K_p = 1.44 \times 10^{-5}$

$$\Delta n = 2 - 4 = -2$$
$$T = 273 + 500 = 773$$
$$K_C = \frac{K_p}{(RT')^{\Delta n}} = \frac{1.44 \times 10^{-5}}{(0.082 \times 773)^{-2}}$$

So, **option 4** is the correct choice.

Q84) EXPLANATION:

$$H_2X \rightleftharpoons H^+ + HX^- (K_{a_1})$$
$$HX^- \rightleftharpoons H^+ + X^{2-} (K_{a_2})$$
$$\text{So} \quad K_{a_1} > K_{a_2}$$

Q85) **Explanation:**

**Hint** Propanoic acid is a weak acid.

**Explanation:**

Let the degree of ionization of propanoic acid be α.

Then, representing propionic acid as HA, we have:

$$HA \ +(.05 - 0.0\alpha) \approx .05H_2O \leftrightarrow \underset{.05\alpha}{H_3O^+} + \underset{.05\alpha}{A^-}$$
$$K_\alpha = \frac{[H_2O^+][A^-]}{[HA]}$$
$$= \frac{(.05\alpha)(.05\alpha)}{0.05} = .05\alpha$$
$$\alpha = \sqrt{\frac{K_\alpha}{.05}} = 1.63 \times 10^{-2}$$

Q86) **Explanation:**

**Hint:** $K_{equ} = \frac{[C][D]}{[A][B]}$

**Explanation:**

**Step 1:** The given reaction is as follows: $A + B \leftrightharpoons C + D$

The equilibrium constant is the ratio of the concentrations of the products to the reactants raised to the stoichiometric coefficients.

For the given reaction, the equilibrium constant is written as: $K_{equ} = \frac{[C][D]}{[A][B]}$

**Step 2:** Let, initially, the concentration of A and B be "a." So, the ICE table can be written as:

| | $A$ | $+$ | $B$ | $\leftrightharpoons$ | $C$ | $+$ | $D$ |
|---|---|---|---|---|---|---|---|
| Initial conc. | a | | a | | - | | - |
| Change in conc. | -x | | -x | | +x | | +x |
| Equilibrium conc. | a-x | | a-x | | x | | x |

It is also given that, at equilibrium, the concentration of D is twice that of A.

So, $2(a-x) = x$ or, $x = \frac{2a}{3}$

**Step 3:** Thus, the value of the equilibrium constant will be:

$$K_{equ} = \frac{x \times x}{(a-x)\times(a-x)}$$

$$K_{equ} = \frac{\frac{2a}{3}\times\frac{2a}{3}}{\frac{a}{3}\times\frac{a}{3}} = 4$$

So, **option 4** is the correct choice.

Q87) **Explanation:**

Hint: $K_C = \frac{[B][C]}{[A]^2}$

Q88) **Explanation:**

Hint: $\Delta n_g = 0$, for $K_p = K_C$

1. The equilibrium constant value only depends on the temperature. It does not depends on the concentration of reactant or product. Hence, first statement is correct.

2. Value of K tells about the extend or reaction, and also tells if the reaction is spontaneous, non spontaneous or in equilibrium.

If K > 1, then reaction is spontaneous.
If K < 1, then the reaction is non-spontaneous.

The second statement is correct.

3. In a homogeneous system, all the reactants and products are in the same phase. The relation between $K_p$ *and* $K_C$ are as follows:
$K_p = K_C(RT)^{\Delta n_g}$

If the value is zero then $K_p$ and $K_C$ value is the same. This is not always true for a homogeneous system $K_p = K_C$.

For example, $H_2(g) + I_2(g) \rightarrow 2HI(g)$, $K_p$ *and* $K_C$ value is not equal.

Thus, option third is the correct answer.

Q89) Hint: $K_{sp}=[Ca^{2+}][F^-]^2$

Step 1:

The reaction is as follows:

$CaF_2=Ca^{+2}+2F^-$

X 2X

The dissociation of NaF is as shown below:

$NaF \rightarrow 0.1Na^+ + 0.1F^-$

Thus total fluoride ion concentration is 2x+0.1 M but 2x<<<0.1.
Hence, 2x is neglected and the total fluoride ion concentration is 0.1 M.
Step 2:
The expression for the solubility product is as follows:

$K_{sp}=[Ca^{2+}][F^-]^2$
$5.3\times10^{-11}=x(0.1)^2$
$x= 5.3\times10^{-9}$ M.

Q90) **Explanation:**

**Hint:** $K_{sp}=\left[Au^{3+}\right]\left[Cl^-\right]^3$

**Explanation:**
Molar solubility $= \frac{1.00\times^{-4} gAuCl_3/303\ g\ AuCl_3\ mol^{-1}}{1\ L}$
$= 3.3\times10^{-7}\ M$

| Reaction | $AuCl_3$ | $\rightleftharpoons$ | $Au^{3+}$ | + | $3Cl^-$ |
|---|---|---|---|---|---|
| Initial Conc. | Solid | | $0\ M$ | | $0\ M$ |
| Change | $-x$ | | $+x$ | | $+3x$ |
| Equilibrium | Solid | | $+x$ | | $+3x$ |
| Answer | | | | | |

$K_{sp}=[Au^{3+}][Cl^-]^3=(x)(3x)^3$
Since $x=3.3\times10^{-7}$
$K_{sp}=27\ S^4=27\times(3.3\times10^{-7})^4=3.2\times10^{-25}$

Hence, the **fourth option** is correct.

# Mitoprep

# Sapno Ki Udaan